Egocentric 3D Hand Pose Estimation

Brett Kinsella
Department of Electrical and Computer Engineering
University of California, San Diego
bkinsella@ucsd.edu

Dave Melkani
Halıcıoğlu Data Science Institute
University of California, San Diego
dmelkani@ucsd.edu

Jeffrey Yang
Department of Computer Science and Engineering
University of California, San Diego
jey013@ucsd.edu

Pu Sun
Department of Electrical and Computer Engineering
University of California, San Diego
p7sun@ucsd.edu

1 Background & Motivation

Estimating 3D hand pose from images is a core problem in computer vision with applications in AR/VR, robotics, and human-computer interaction. Egocentric settings are particularly important because they capture hands during natural interactions and enable systems to understand fine-grained human actions. This problem is challenging when generalizing to real-world egocentric images. Hands are often occluded by objects, appear at low resolution, and exhibit strong perspective distortion due to their proximity to the camera. In addition, large-scale datasets with accurate 3D annotations are limited outside controlled environments, which restricts model generalization. Real-time applications also impose strict latency constraints, requiring models to balance accuracy with computational efficiency.

Recent work such as WildHands (1) addresses these issues by incorporating camera-aware representations and leveraging auxiliary supervision from in-the-wild data, improving generalization across diverse datasets. This project aims to reproduce the WildHands method and evaluate the impact of different backbone architectures. The goal is to analyze how backbone choice affects performance and generalization in egocentric 3D hand pose estimation. If time permits, we will also explore adapting the method to the HOT3D (2) dataset, which requires handling fisheye camera models instead of standard pinhole assumptions.

2 Related Work

Prior research relevant to our project can be grouped into three main categories: unconstrained egocentric environment adaptation, the evolution of backbone architectures for feature extraction, and perspective-aware 3D representations.

Unconstrained egocentric environments: Early 3D hand pose estimation methods typically relied on data captured in controlled labs, which do not translate well to everyday interactions where hands are heavily occluded by objects and distorted by the camera's close perspective. Recent works like WildHands (1) address this gap by using camera-aware representations and learning from 2D auxiliary data (e.g., segmentation masks) rather than relying solely on 3D ground truth. Our project is situated directly within this problem setting, as we use the WildHands framework as our baseline to understand how models handle messy, real-world images. Accurate pose estimation in these settings also serves as a prerequisite for downstream generative tasks: MEgoHand (3) demonstrates that egocentric hand-object motion generation (which synthesizes physically plausible future hand trajectories from RGB, text, and an initial MANO pose) depends critically on robust 3D hand understanding under the same challenging conditions of occlusion and perspective distortion that our work targets.

Neural network backbones: Many standard approaches still use Convolutional Neural Networks (CNNs) such as ResNet-50 (4) for 3D hand pose estimation. CNNs are fast and efficient for real-time tracking but struggle with heavy occlusions because they primarily focus on local pixel-level features. To address this, recent state-of-the-art models like HaMeR (5) have shifted to massive Vision Transformers (ViTs) which excel at capturing global context to infer the location of hidden joints, at the cost of significantly greater computational demand. The importance of global spatial reasoning for occluded egocentric hands is further reinforced by MEgoHand (3), which employs a vision-language model for intent and spatial understanding, suggesting that transformer-based global context is valuable not only for estimation but for any task requiring holistic hand-scene understanding. Additionally, newer architectures like DF-Mamba (6) are now exploring state-space models to combine the speed of CNNs with the global understanding of ViTs.

Perspective-aware 3D representations: WildHands's KPE uses camera intrinsics and sinusoidal functions to mitigate perspective distortion from hand proximity. This parallels PAInpainter (7), which demonstrates that perspective-aware reasoning is essential for maintaining 3D consistency when camera geometry is non-trivial. While PAInpainter addresses scene-level 3D Gaussian inpainting rather than pose estimation, its core insight that naively ignoring camera perspective degrades 3D coherence applies equally to our setting.

3 High-level Methodology

At a high level, our project aims to reproduce the methodology developed in the WildHands paper (1) and extend its findings by testing a variety of different neural network backbones and time permitting, modify the networks architecture for compatibility with Meta's Project Aria HOT3D dataset (2) in order to evaluate the models ability to generalize to fisheye-distorted egocentric settings.

The WildHands model works by first cropping input images around the hand before using a ResNet-50 neural network (8), the model's backbone, to extract features from the cropped image. These hand features are augmented with intrinsics-aware positional encodings (KPE) that supply the network with the hand crop's angular location within the camera's field of view. This information is then further processed and sent to the hand decoder which iteratively predicts the MANO (9) parameters (shape, articulation and global pose) and weak perspective camera parameters in order to create the 3D mesh that estimates the hand pose.

Rather than keeping the default ResNet-50 backbone, we will swap and evaluate CNNs, ViTs, and Mamba-based architectures to analyze their accuracy and efficiency trade-offs in egocentric pose estimation. More specifically, we will test the effect of swapping the default ResNet-50 backbone with MoblieNet-V3-Large (10), MoblieVIT-S (11), ResNet-101 (8), ConvNeXt-L (12) and DF-Mamba (6) backbones. This allows us to situate our work in the current literature by testing if we can achieve the occlusion-handling benefits of modern architectures without sacrificing efficiency in egocentric applications.

The authors of the original WildHands paper trained the model using an egocentric split of the ARCTIC (13), AssemblyHands (14), Epic-Kitchens (15) and Ego4D (16) datasets, and conducted zero-shot evaluation on the H2O (17), AssemblyHands, Epic-HandKps (1) and Ego-Exo4D (18) datasets. We will mirror this training and evaluation methodology in order to offer a direct comparison to the original results, and time permitting, extend the results by using Meta's Project Aria Hot3D dataset for zero-shot evaluation as well. Evaluating the model on the Hot3D dataset would allow us to better understand the model's ability to generalize to fisheye-distorted egocentric settings and may offer unique insight into how current model architectures perform on state of the art platforms like Meta's Project Aria smart glasses (19), which collected samples for the HOT3D dataset. However, we would like to reiterate that this dataset may not appear in the final report due to the heavy modifications to each sample that are required for compatibility with the WildHands model, such as perspective cropping and attaining 2D projections from the provided 3D data.

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